MILITARY SPECIFICATION

CUTTER, DOUGH, HAND ROLLING

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a hand-rolling cutter for dividing dough into equal sizes.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-Z-363	-	Zinc-Base Alloy; Die Castings
PPP-B-636	-	Box, Fiberboard

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-130	-	Identification Marking of U.S. Military Property
MIL-STD-147	-	Palletized and Containerized Unit Loads, 40" x 48" Pallets, Skids, Runners, or Pallet-Type Base

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(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on the date of invitation for bid, or request for proposal shall apply.

* NATIONAL SANITATION FOUNDATION (NSF)

Standard No. 2 - Food Service Equipment.
Listing of Food Service Equipment.

(Application for copies should be addressed to the National Sanitation Foundation, P. O. Box 1[illegible]8, Ann Arbor, Michigan 48106).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

Specification A167-63 - Corrosion-Resisting Chromium-Nickel Steel Plate, Sheet, and Strip.
Specification A276-65 - Hot-Rolled and Cold Finish Stainless and Heat-Resisting Steel Bars.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103).

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 202 Union Station, 516 W. Jackson Blvd., Chicago, Illinois 60606).

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC.

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Attn: Tariff Order Section, 1616 P Street, N.W., Washington, D.C. 20036).

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(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Standard product.- Dough cutters delivered under the specification shall be the manufacturer's current commercial product except for any changes necessary to comply with specification requirement.

* 3.2 Standards.- Dough cutters covered by this specification shall conform to the applicable requirements of NSF Standard No. 2.

3.2.1 NSF certification.- The supplier shall submit satisfactory evidence to the contracting officer or his authorized representative that the dough cutters he proposes to furnish under this specification meet the applicable requirements of NSF Standard No. 2 as follows:

(1) A listing in the current edition of the NSF "Listing of Food Service Equipment" or display of the NSF seal on the finished dough cutters shall be submitted prior to the approval of the preproduction sample, if one is submitted, or prior to the approval of the first shipment, or

* (2) A certification for the dough cutters issued by NSF under their special one-time contract evaluation/certification service shall be submitted prior to the approval of the preproduction sample, if one is submitted, or prior to the approval of the first shipment, or

(3) A certified test report from an independent testing laboratory acceptable to the medical department of the service for which the dough cutters are being procured, indicating that the dough cutters have been tested and conform to NSF Standard No. 2 shall be submitted prior to the approval of the preproduction sample, if one is submitted, or prior to commencing production.

3.3 Materials.-

3.3.1 Corrosion-resisting steel.-

* 3.3.1.1 Bar.- Corrosion-resisting steel bar shall conform to the requirements for type 302 or 304, round form of ASTM A276-65.

* 3.3.1.2 Plate, sheet, or strip.- Corrosion-resisting steel plate, sheet, and strip shall conform to the requirements for type 301, 302, or 304 of ASTM A167-63.

3.3.2 Die castings.- Die castings shall be zinc based alloys conforming to the requirements of QQ-Z-363.

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3.3.3 Plastic.- Plastic, when used for handles shall be the manufacturer's standard commercial product.

3.4 Design.- The dough cutter shall be the hand rolling type and shall consist of 18 removable dough cutting blades mounted in two circular end plates or castings spaced 5-3/4 ± 1/4 inch apart; a handle rod passing through the centers of the end plates or casting; and two wood or plastic handles mounted on the handle rod. The cutting blades shall be uniformly spaced around the circumference of the end plates so that when all cutting blades are in position one revolution of the dough cutter will produce 18 uniform sized dough pieces measuring 5-3/4 by 1-3/8 ± 1/4 inch (see 4.4.1).

3.5 Construction.-

3.5.1 End-plates.- The end-plates shall be formed from the corrosion-resisting steel specified in 3.3.1.2 or die cast of the zinc alloy specified in 3.3.2. The plates shall be 7-3/8 inches diameter, plus or minus 1 inch, and not less than 0.093 inch thick. The plates shall be uniformly notched or grooved to receive and rigidly support the blades and shall contain a center hole to receive the handle rod. End-plates shall be provided with not less than four bolt holes to receive bolts for fastening the end-plates together and for holding the blades firmly in place. Removal and replacement of the blades shall not require the use of special tools.

3.5.2 Blades.- The 18 blades shall be fabricated of corrosion-resisting steel specified in 3.3.1.2. Each blade shall have a thickness of 0.036 inch, plus or minus 0.005 inch, and a width of 1-5/16 inches, plus or minus 1/8 inch. Blades shall have sharpened cutting edges. Blade length shall be such that when inserted in position on the end-plates, the distance between the end-plates is suitable for obtaining dough pieces conforming to the size specified in 3.4. The cutting edge of each blade shall project 3/8 inch, plus or minus 1/16 inch, above the diameter of the end-plates.

3.5.3 Handles and handle rod.- The handle rod shall be of corrosion-resisting steel specified in 3.3.1.1 and shall be 1/2 inch diameter, plus or minus 1/16 inch, and 14-3/4 inches long, plus or minus 1 inch, with threaded ends. Handles shall be of hardwood or of the plastic specified in 3.3.3 and shall be of a diameter and shape suitable for a comfortable handgrip. The handles shall be 3-3/4 inches long plus or minus 1/2 inch. The handles shall fit over the handle rod and the handles shall be held in place by corrosion-resisting steel nuts fitted to the rod. The handles shall rotate on the handle rod or the handle rod shall rotate in the end plates.

3.6 Marking for identification.- Each dough cutter shall be permanently and legibly marked with the manufacturer's name, trade name, or trade mark of such character as to be identifiable with the manufacturer. Marking shall be applied in accordance with MIL-STD-130.

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3.7 Finish.- Dough cutters shall be finished to conform to NSF Standard No. 2. Plastic handles, when used shall have a gloss finish obtained during the molding process.

3.8 Workmanship.- The finished dough cutter shall be clean and free of rust, slivers, die marks, deformities, and fractures. All sharp edges, except cutting edges, shall be removed.

* 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Component and material inspection.- In accordance with 4.1, components and materials shall be inspected and tested in accordance with all requirements of referenced specifications, standards, and drawings unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.2.2 End item inspection.- The inspection lot shall consist of all dough cutters offered for inspection at one time. The sample unit for this inspection shall be one completely fabricated and assembled dough cutter with 18 cutting blades in place.

4.2.2.1 Visual examination.- Examination shall be made of the dough cutters for the defects listed in table I. The inspection level shall be level II with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

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TABLE I.- Classification of defects

Examine	Defect	Classification Major	Minor
Finish	Dirty		X
	Plastic handles, when used, do not have a gloss finish obtained by the molding process	X	
Design	Not manufacturer's standard product, except as otherwise specified	X	
	Has more or less than 18 cutting blades	X	
	Cutting blades not removable or not uniformly spaced	X	
	Handles or handle rod do not rotate	X	
	Handles not fastened to handle rod with corrosion-resisting steel nuts	X	
Construction and workmanship	Slivers, deformities, fractures	X	
	Sharp edges (except cutting edges) not removed	X	
	Die marks not removed		X
Marking for identification	Missing, illegible		X

4.2.2.2 Dimensional examination.- An examination shall be made to determine compliance with dimensions specified. Any nonconformance shall be considered a defect. The inspection level shall be S-2 with an AQL of 4.0 defects, expressed in terms of defects per hundred units.

4.2.2.3 Testing of the end item.- One unit of production shall be tested as specified in 4.4.1. Failure to pass this test shall be cause for rejection of the lot.

4.2.3 Examination of preparation for delivery.- An examination shall be made to determine that packaging, packing, and marking as required by section 5 of this specification are complied with. Defects shall be as specified in table II. The lot shall be the number of containers offered for inspection at one time. The sample unit shall be one container fully prepared for delivery. The inspection level shall be S-2 with an AQL of 4.0 expressed in terms of defects per hundred units.

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TABLE II.- Examination of preparation for delivery

Examine	Defect
Marking (exterior and interior)	Missing; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Components missing or damaged
Weight	Net weight exceeds requirements

4.2.3.1 Examination of shipping containers.- When shipping containers are required to be in accordance with PPP-B-636, examination for defects in closure, waterproofing, and strapping shall be in accordance with PPP-B-636.

4.2.3.2 Examination of palletization.- An examination shall be made to determine that palletization is in compliance with section 5 of this specification. Defects shall be as specified in table III. The sample unit for this examination shall be one palletized unit load ready for shipment. The lot shall be the number of palletized unit loads submitted for inspection at one time. The inspection level shall be S-1 with an AQL of 6.5 expressed in terms of defects per hundred units.

TABLE III.- Examination of palletization

Examination	Defect
Finished dimension	Length, width, or height exceeds specified maximum requirement
Palletization	Not as specified Pallet pattern not as specified Interlocking of loads not as specified Load not bonded with required straps specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

4.3 Code and standards compliance.- Proof of compliance with the requirements of 3.2 shall be submitted to the Government representative.

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4.4 Test.-

4.4.1 Functional test.- The dough cutter with 18 cutter blades in place shall be rolled over a piece of dough of sufficient mass, length, and width to produce 18 pieces of dough. The dough pieces shall then be measured to determine compliance with the dough size requirements of 3.4. Any nonconformance with specified requirements shall constitute failure of this test.

5. PREPARATION FOR DELIVERY

5.1 Packing.- Packing shall be level A or C as specified (see 6.2).

5.1.1 Level A.- Each dough cutter shall be packaged in a snug-fitting fiberboard box conforming to style RSC, type CF, variety optional, or type SF, class domestic of PPP-B-636. Box closure shall be in accordance with the appendix of PPP-B-636.

5.1.2 Level C.- Each dough cutter shall be packaged to afford adequate protection against damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets this requirement.

5.2 Packing.- Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A.- Dough cutters, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, V2s of PPP-B-636. Closure, waterproofing by means of tape, and reinforcing with flat strapping or tape banding shall be in accordance with the appendix of PPP-B-636. Weight of contents of each shipping container shall not exceed 65 pounds.

5.2.2 Level B.- Dough cutters, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF, variety optional, or type SF, class domestic, grade 275 of PPP-B-636. Closure shall be in accordance with method II of the appendix of PPP-B-636. Weight of contents of each shipping container shall not exceed 65 pounds.

5.2.2.1 When specified (see 6.2) the shipping container shall be V3c, V3s, or V4s, fabricated in accordance with PPP-B-636 and closed in accordance with the appendix thereto.

5.2.3 Level C.- Dough cutters, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

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5.3 Palletization.- Unless otherwise specified (see 6.2) dough cutters packaged as specified shall be palletized in accordance with Load Type I of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

5.4 Marking.- In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

* 6. NOTES

6.1 Intended use.- The dough cutter covered by this specification is intended for use in Military kitchens, galleys, and bakeries to divide dough into equal parts for buns and rolls.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(c) When weather-resistant class fiberboard shipping containers are required for level B shipment (see 5.2.2.1).
(d) When palletization is not required (see 5.3).

6.3 The margins of this specification are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - SA
Air Force - 84

Preparing activity:

Army - GL

Project No. 7330-0278

Review activities:

Army - MD
Navy - MC

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

Cutter, Dough, Hand Rolling MIL-C-40616B

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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